Q1:JUN 2008 P2

8 **Fig. 8.1** shows a logic circuit.

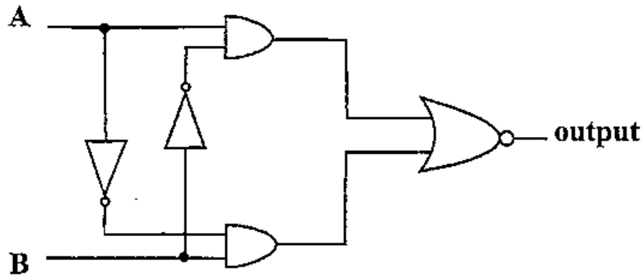


Fig. 8.1

(i) Draw a truth table for the circuit. [4]

(ii) Identify a single gate which is equivalent to the circuit.

____ [1]

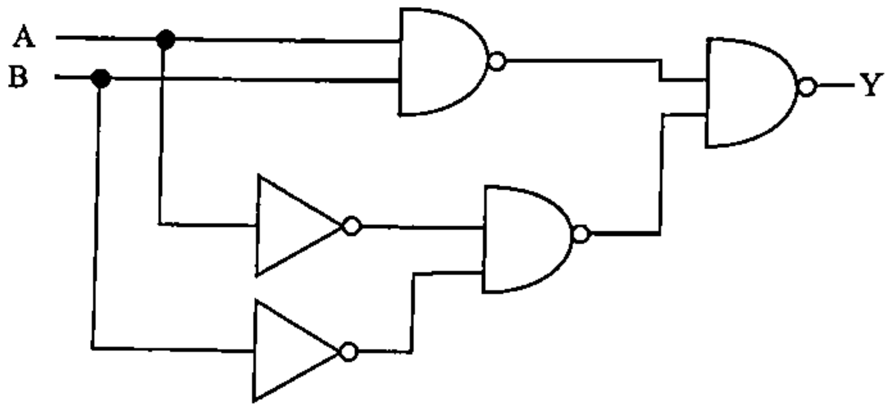
Q2:NOV 2013 P2

7 **(a)** Define a logic gate.

____

____ [1]

(b) **Fig. 7.1** shows a combination of logic gates.



Q3:NOV 2015 P2

7 **(a)** Define *a logic gate.*

____

____

____ [1]

(b) **Fig.7.1** shows a combination of logic gates.

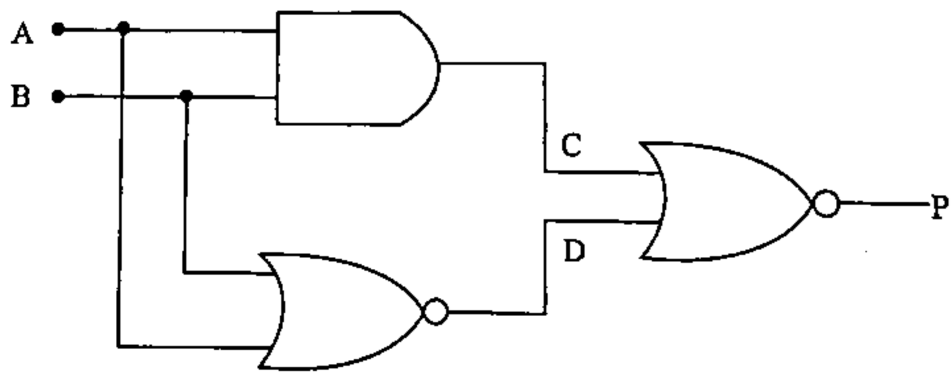


Fig.7.1

(i) Construct a truth table for the combination.

(ii) State the logical function for the combination.

____ [4]

Q4:NOV 2018 P2

7 **Fig. 7.1** shows a combination of logic gates.

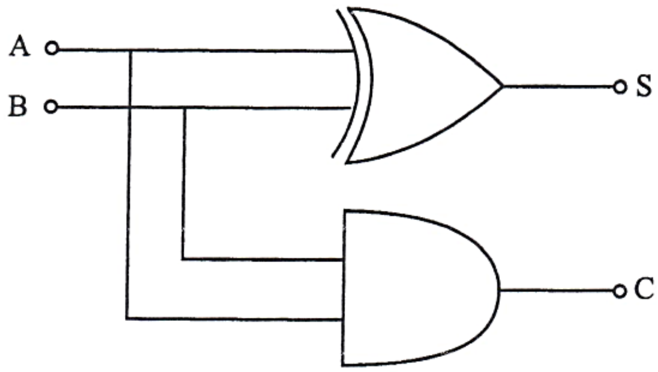


Fig. 7.1

(a) Name the logic gates used.

____ [1]

(b) Complete the truth table for the circuit in **Fig.7.1**.

A	B	C	S
0	0		
0	1		
1	0		
1	1		

[2]

Q5:JUN 2007 P3

5 (a) (i) Draw the symbol of a 2-input **OR** gate.

(ii) Write down its truth table. [2]

(b) (i) Draw a truth table for the circuit shown in **Fig. 5.1**.

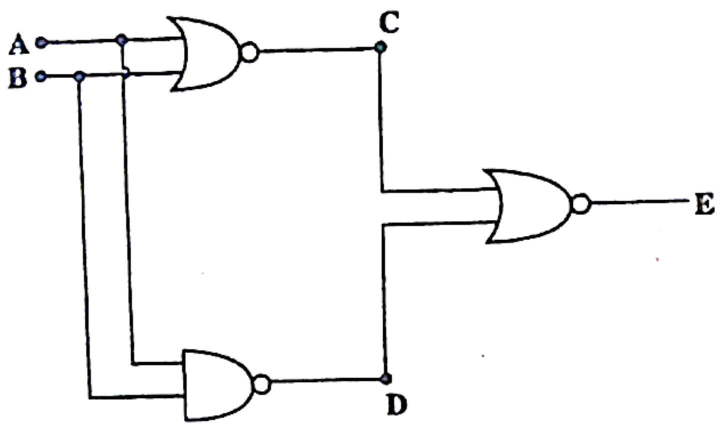


Fig. 5.1

(ii) State the name given to a gate which behaves as the circuit in **b(i)**. [4]

(c) **Table 5.1** shows the characteristics of an LDR.

Table 5.1

Resistance/kΩ	0.1	1.0	10	100	1 000
Intensity/Wm^{-2}	100	10	1.0	0.1	0.01

The LDR is incorporated into the circuit of **Fig. 5.2**.

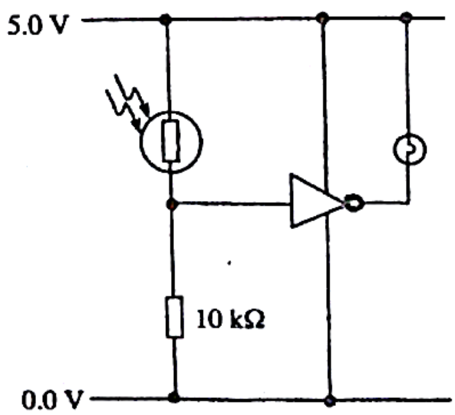


Fig. 5.2

(i) Explain how the circuit works.

(ii) Deduce from **Table 5.1** the range of the light intensity in which the bulb lights.

(iii) Suppose the bulb is placed close to the LDR and the system is initially in the dark . Explain what happens when a torch is briefly shone on the LDR.

(iv) Explain one application of the circuit in **Fig. 5.2.** [9]

(d) The circuit of **Fig. 5.2** was modified as in **Fig. 5.3.**

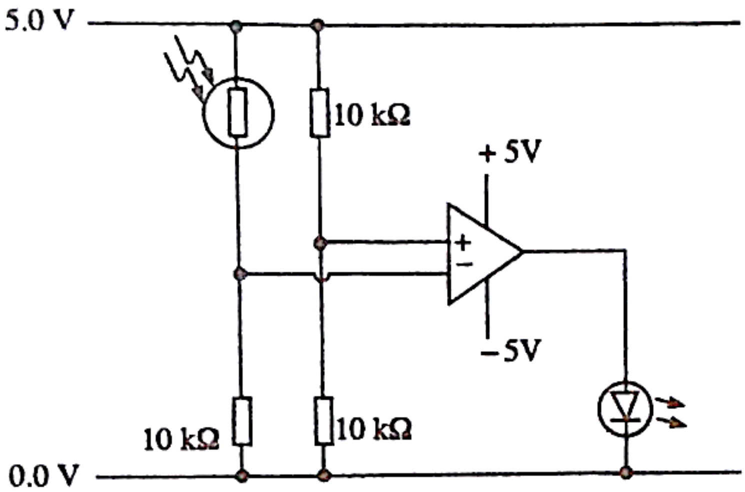


Fig. 5.3

(i) Identify the circuit.

(ii) Calculate the potential at A when the light intensity on the LDR is 10 Wm^{-2}.

(iii) Deduce the conditions under which the LED will light. [5]

Q6:NOV 2006 P3

5 (c) **Fig. 5.1** shows a combination of logic gates.

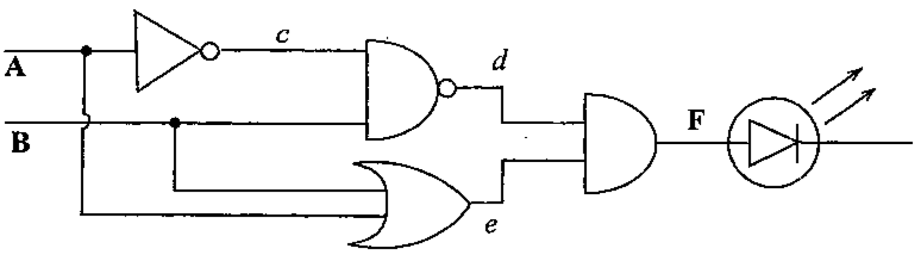


Fig. 5.1

Show that the output of the L.E.D is independent of the state of B. [4]

(d) **Fig. 5.2** shows a two way switch.

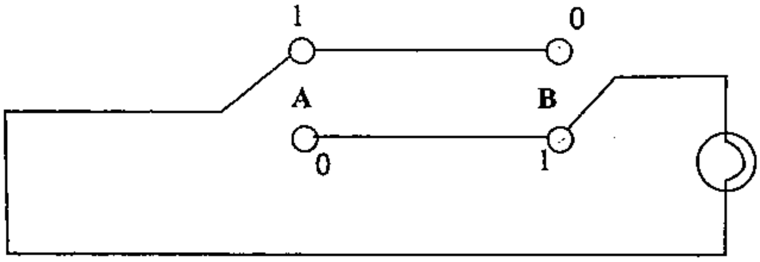


Fig. 5.2

(i) Draw a truth table to show how this switch operates.

(ii) Give **one** gate that is represented by this circuit. [3]

(e) **(i)** Name **one** electronic device and **one** analogue device that perform the same functions.

(ii) Give **two** advantages of the electronic device. [4]

Q7:NOV 2009 P5

5 **(c)** **Fig. 5.2** shows three logic gates connected together.

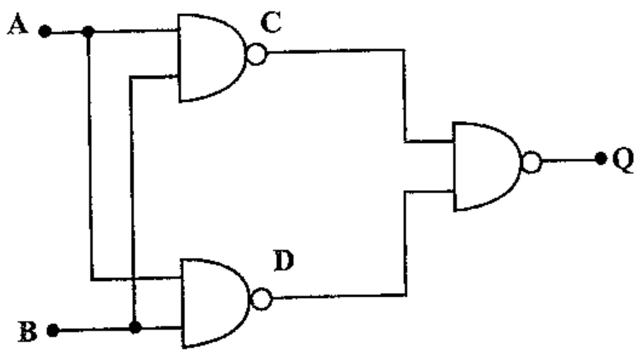


Fig. 5.2

(i) Give the name of the logic gates used.

(ii) Draw the truth-table for the circuit shown.

(iii) Write down the name of a single logic gate that performs the same function as the circuit in **Fig. 5.2**. [5]

Q8:NOV 2012 P5

4 (a) (i) Write down the truth table for a two input Exclusive OR gate.

(ii) Construct a combination of NAND gates to form an Exclusive OR gate. [5]

(b) Draw the truth table for the logic circuit shown in Fig.4.1. [6]

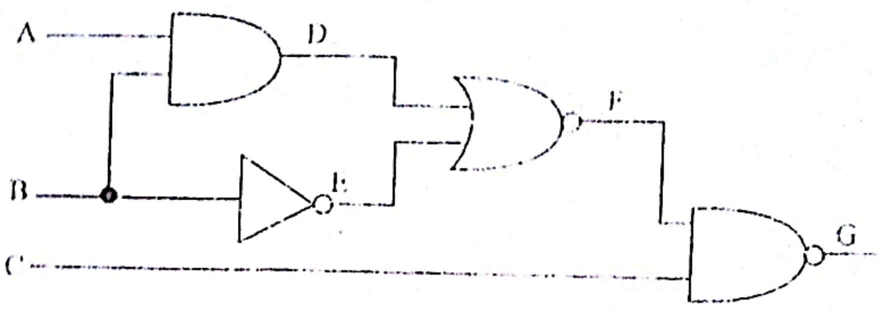


Fig. 4.1

(c) Receiving information of events in the world such as sporting news and natural disasters is almost instantaneous.

State **three** aspects of modern communication which have made this possible. [3]

Q9:JUN 2010 P5

5 (a) Fig. 5.1 shows an alarm circuit to wake up school children in the morning.

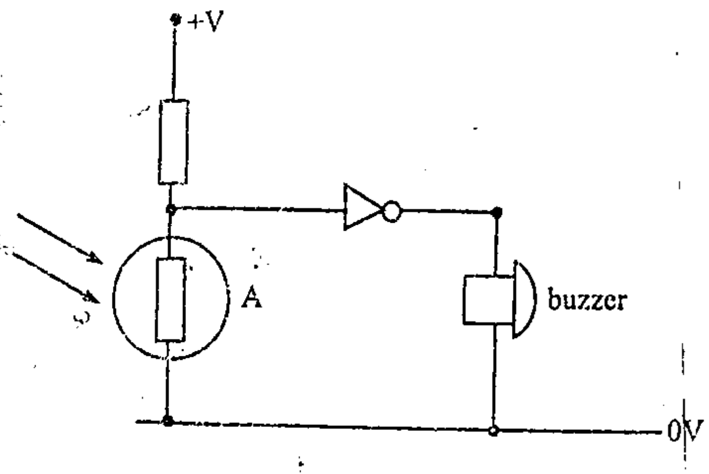


Fig. 5.1 .

(i) State the name of component A.

(ii) Describe and explain how the circuit operates.

(iii) Draw a truth table for the circuit. [7]

(b) Briefly outline **three** positive and **two** negative impacts of modern electronic technology on society. [5]

Q10:JUN 2011 P5

5 (a) Describe the use of an Operational Amplifier as a summing amplifier in the inverting mode. [3]

(b) The circuit shown in **Fig. 5.1** is a simple combination locking system which will allow access to a bank safe if the correct combination is set to A, B, C and D. If there is any wrong combination the alarm will sound.

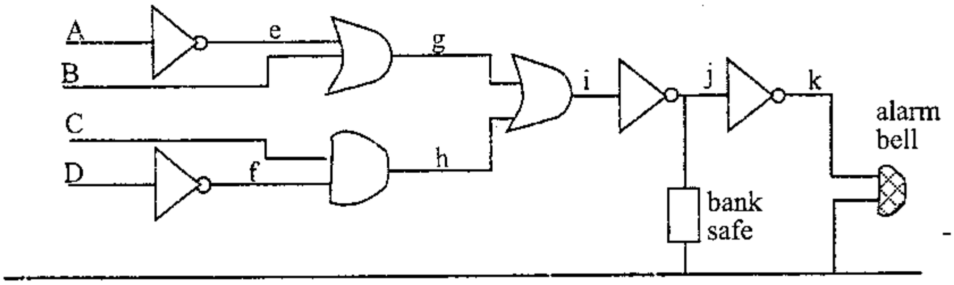


Fig. 5.1

(i) Work out the correct codes to open the bank safe.

(ii) Give **one** advantage of this method.

[5]

(c) Describe the three parts of a radio system in terms of the *input*, *electronic circuit* and the *output device*. [3]

(d) State **one** negative impact of electronics in society. [1]

Q11:JUN 2018 P5

5 **(a) Fig.5.1 represents a vehicle alarm system.**

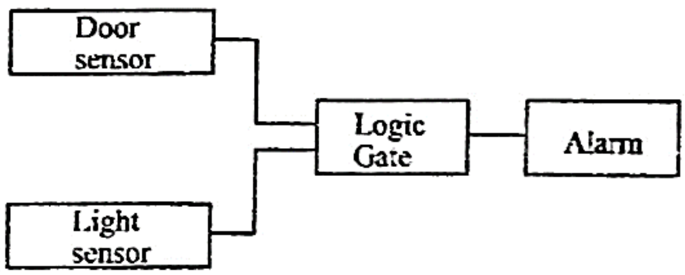


Fig.5.1

The system should be such that the alarm sounds when either the door is open or the light is on or both.

(a) Identify a suitable 1. light sensor,

2. logic gate,

3. output transducer.

[3]

Q12:JUN 2019 P3

3 **(d)** **Fig. 3.4** shows a combination of logic gates.

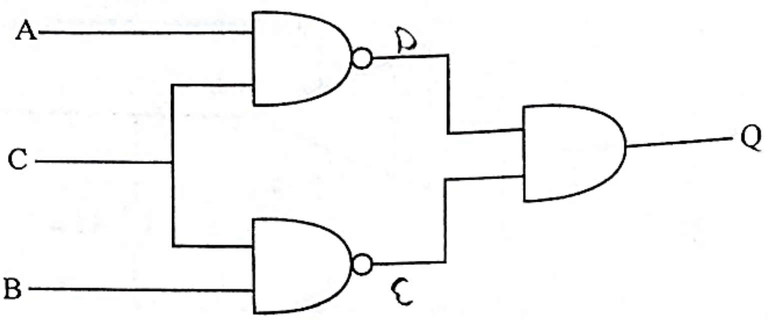


Fig. 3.4

Input C is maintained at logic level 1.

(i) Construct the truth table for the combination.

(ii) Hence deduce the single logic gate that is equivalent to the combination.

[4]